

1606

Minerals, Industries and Transport of India

1 to 2 questions a paper, and the highway corridor block alone has appeared in five separate papers.

Minerals feed the Rajasthan Geography slot as well. If you are short of time in this subject, do this chapter first and the highway section within it first of all.

This is the highest-yield chapter in Indian Geography, and it contains the single most repeated geography topic in the whole RAS paper - the national highway corridors and their crossing at Jhansi, asked in 2013, 2016, 2021, 2023 and 2024. Everything else here is mineral-to-place and plant-to-collaborator matching.

Minerals, Industries and Transport

Iron ore belts

- Odisha-Jharkhand
- Durg-Bastar-Chandrapur
- Bellary-Chitradurga
- Maharashtra-Goa

Other metals

- Manganese-Balaghat
- Bauxite-Panchpatmali
- Copper-Malanjkhand
- Lead-zinc-Rampura Agucha

Energy

- Jharia coking coal
- Neyveli lignite
- Mumbai High
- Rawatbhata
- Kudankulam

Steel plants

- Jamshedpur-Tata
- Bhilai-USSR
- Rourkela-W Germany
- Durgapur-UK
- Vizag-shore

Industrial regions

- Hugli oldest
- Mumbai-Pune
- Chotanagpur
- Gujarat
- Bengaluru-TN

Highways

- Golden Quadrilateral
- North-South NH-44
- East-West NH-27
- Cross at Jhansi

Rail and water

- 19 zones
- DFC East and West
- NW-1 Ganga
- NW-2 Brahmaputra

Ports

- 12 major ports
- JNPT container
- Kolkata riverine
- Vadhavan new

Metallic minerals – the belts and who leads

Mineral questions come in one shape: a mine or a belt on one side, a state on the other. So learn the place name first and the geology never. One caution – for several minerals the largest producer and the largest reserve holder are different states, and RPSC builds assertion-reason and NOT-matched questions on exactly that gap.

Mineral – belt or mine – state

Mineral	Belt / mine	State
Iron ore	Badampahar, Gua, Noamundi	Odisha and Jharkhand
Iron ore	Bailadila, Dalli-Rajhara	Chhattisgarh (Durg-Bastar-Chandrapur belt)
Iron ore	Kudremukh	Karnataka (Bellary-Chitradurga-Tumakuru belt)
Manganese	Balaghat	Madhya Pradesh
Manganese	Nagpur-Bhandara	Maharashtra
Copper	Malanjkhand (Balaghat)	Madhya Pradesh
Copper	Khetri	Rajasthan
Copper	Singhbhum	Jharkhand
Bauxite	Panchpatmali (Koraput), Kalahandi	Odisha
Bauxite	Lohardaga	Jharkhand
Lead-zinc	Rampura Agucha, Zawar, Rajpura-Dariba	Rajasthan
Gold	Hutti and Kolar	Karnataka
Chromite	Sukinda valley	Odisha

- Odisha is the largest producer of iron ore in India.
- Bailadila haematite (Dantewada, Chhattisgarh) is exported to Japan through Visakhapatnam.

- Kudremukh magnetite (Karnataka) moved out through New Mangalore.
- Manganese – Madhya Pradesh leads production (about 31 per cent as per the Indian Bureau of Mines); Odisha holds the largest reserves (about 34 per cent).
- Copper – Madhya Pradesh is the largest producer; Rajasthan holds over half the country's copper reserves.
- Lead-zinc – almost the entire Indian output is from Rajasthan, worked by Hindustan Zinc.
- Aluminium – Odisha leads; NALCO refines at Damanjodi and smelts at Angul, BALCO is at Korba, HINDALCO at Renukoot.
- Kolar Gold Fields closed in 2001; Hutti still works, and Ramagiri is in Andhra Pradesh.
- Three copper belts, west to east: Khetri (Rajasthan), Malanjkhand (MP), Singhbhum (Jharkhand).

TRAP

Manganese is the classic assertion-reason trap. 'Odisha is the leading producer of manganese' is FALSE – Madhya Pradesh is. But 'Odisha holds the largest manganese reserves' is TRUE. Producer MP, reserves Odisha. Copper works the same way: producer MP, reserves Rajasthan.

Non-metallic and atomic minerals

A short section but a dense one – almost every entry is a one-line pair, and several of them are in Rajasthan, which is why RPSC keeps coming back to them. Note the split that runs through this section too: Rajasthan is the largest producer of limestone, but Karnataka holds the largest reserves.

Non-metallic and atomic minerals

Mineral	Chief location	State / note
Limestone	Widespread	Rajasthan largest producer (about 22%); Karnataka largest reserves
Rock phosphate	Jhamaikotra	Rajasthan (Udaipur) - India's chief mine
Gypsum	Nagaur, Bikaner	Rajasthan
Mica	Koderma-Giridih	Jharkhand; also Nellore (AP) and Bhilwara
Diamond	Panna	Madhya Pradesh
Uranium	Jaduguda	Jharkhand - India's first uranium mine
Uranium	Tummalapalle	Andhra Pradesh
Chromite	Sukinda	Odisha - nearly all of India's output

Energy resources - coal, oil, gas and nuclear

Coal first. India's coal is overwhelmingly Gondwana coal, found in the peninsular river valleys; Tertiary coal is younger, poorer and sits in the north-east. The examiner's favourite pair here is 'oldest coalfield' and 'coking coal coalfield' - they are two different fields. Then move to oil, where Rajasthan has a big stake.

- About 96 per cent of India's coal reserves are Gondwana coal.
- Jharia is the largest coalfield and the chief source of prime coking coal.
- Raniganj, where mining began in 1774, is the oldest coalfield.
- Other fields: Bokaro, Karanpura, Korba, Singrauli, Talcher.
- Jharkhand has the largest coal reserves; Odisha has led production in recent years.
- Lignite - Neyveli (Tamil Nadu) is the largest field; Kapurdi-Jalipa and Barsingsar are in Rajasthan; Panandhro in Gujarat.

- Mumbai High, found 1974 and drilled by the rig Sagar Samrat, is India's largest oil field.
- Rajasthan's Barmer basin - the Mangala field, discovered 2004 - makes it the leading onshore crude producer.
- Gujarat has Ankleshwar and Kalol; Assam has Digboi, Naharkatiya and Moran.
- Digboi, 1901, is India's oldest refinery; Jamnagar in Gujarat is the world's largest refining complex.
- The HPCL Rajasthan Refinery is at Pachpadra in Balotra, erstwhile Barmer district.
- Natural gas moves through GAIL's Hazira-Vijaipur-Jagdishpur (HVJ) pipeline.
- Hydel power comes from the multipurpose dams of the previous chapter; among renewables, Bhadla in Rajasthan is among the world's largest solar parks.

ASKED IN RAS

Rajasthan's fossil fuels have been asked twice over. RAS Pre 2024 - which lignite field lies in Rajasthan? Answer: Kapurdi-Jalipa (Barmer); Barsingsar and Palana are the other names. RAS Pre 2013 and 2018 - the Mangala oil field lies in which basin? Answer: the Barmer basin, with Bhagyam and Aishwariya as the companion fields.

Nuclear power plant - state - identifier

Plant	State	Identifier
Tarapur	Maharashtra	Oldest, commissioned 1969
Rawatbhata	Rajasthan	Built with Canadian help
Kalpakkam	Tamil Nadu	Fast breeder programme site
Kudankulam	Tamil Nadu	Largest, Russian-aided
Kakrapar	Gujarat	India's first indigenous 700 MWe PHWR
Narora	Uttar Pradesh	On the Ganga canal

Plant	State	Identifier
Kaiga	Karnataka	Western Ghats site

TRAP

Two planted errors sit together here. 'Rajasthan has the largest coal reserves' is false - Rajasthan has lignite, Jharkhand has the coal. And Neyveli is a lignite-thermal complex, not a nuclear station; it is the standard wrong option in 'which is NOT a nuclear power station'.

Iron and steel, and the other big industries

Steel plants are asked in exactly one way: which country helped build which plant. Learn the four SAIL collaborations as a block, then add the two that stand apart - Tata's private plant at Jamshedpur on the Subarnarekha, India's first modern steel plant, and Visakhapatnam, the only shore-based integrated plant.

Steel plant - state - collaboration

Plant	State	Built with
Jamshedpur (TISCO)	Jharkhand	Private, Tata - founded 1907, production from 1912
Bhilai	Chhattisgarh	USSR
Bokaro	Jharkhand	USSR
Rourkela	Odisha	West Germany
Durgapur	West Bengal	United Kingdom
Visakhapatnam (RINL)	Andhra Pradesh	Only shore-based integrated plant

- Order of production: Jamshedpur 1912, Bhilai 1959, Bokaro early 1970s, Visakhapatnam 1992.
- The first cotton textile mill was set up at Mumbai in 1854; the first jute mill at Rishra on the Hooghly in 1855.
- Ahmedabad is the 'Manchester of India'; Coimbatore the 'Manchester of South India'.
- Sugar - Uttar Pradesh has the most mills and the largest cane area; Maharashtra has higher recovery.
- Cement - the first plant was set up at Chennai in 1904; India is the world's second largest cement producer.
- Sindri, 1951, was India's first large fertiliser plant.
- Chennai is the 'Detroit of India'; other auto hubs are Gurugram-Manesar, Pune-Chakan, Sanand and Pithampur.
- Bengaluru is the 'Silicon Valley of India'.

YAAD RAKHO

Two Russians, one German, one Briton: Bhilai and Bokaro from the USSR, Rourkela from West Germany, Durgapur from the United Kingdom. The D plant is NOT Deutschland - that is exactly the slip the option list is waiting for.

Industrial regions and corridors

India is usually described as having eight major industrial regions. Learn each with its signature industry, because the matching question uses the industry, not the geography.

Industrial region - characteristic industry

Region	Signature industry
Hugli (the oldest)	Jute
Mumbai-Pune	Cotton textiles, engineering, films
Chotanagpur	Iron and steel, heavy metallurgy
Gujarat	Cotton textiles and petrochemicals
Vishakhapatnam-Guntur	Shipbuilding and petroleum refining
Bengaluru-Tamil Nadu	Electronics, engineering, textiles
Gurugram-Delhi-Meerut	Light engineering, electronics, automobiles
Kollam-Thiruvananthapuram	Coir, rubber, cashew, sugar, aluminium

- Dedicated Freight Corridors: Western DFC from Dadri (UP) to JNPT Mumbai, about 1,506 km, with Japanese assistance.
- Eastern DFC from Sahnewal (Ludhiana) to Sonnagar (Bihar), about 1,337 km, with World Bank assistance.
- The Delhi-Mumbai Industrial Corridor runs along the Western DFC and is Japan-aided.
- About 39 per cent of the DMIC alignment lies in Rajasthan - the state's biggest industrial hook.
- Other corridors: Chennai-Bengaluru, Amritsar-Kolkata, Vizag-Chennai.
- Sagarmala (2015) is port-led development; Make in India is 2014; PM Gati Shakti is 2021 for multimodal connectivity.
- Two planted errors: Kollam-Thiruvananthapuram rests on coir, rubber and cashew, NOT iron and steel; and Chotanagpur is Jharkhand, not Chhattisgarh.

National highways and the corridors - the most repeated topic in

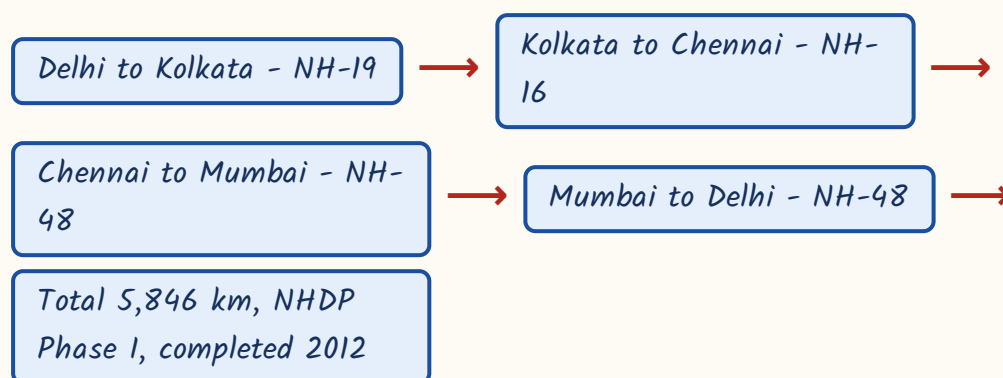
RAS geography

Give this section a whole revision slot to itself. RPSC has asked from it in 2013, 2016, 2021, 2023 and 2024 - five papers. Nothing else in geography

comes close. The National Highways Development Project was built in phases. Phase I was the Golden Quadrilateral joining the four metros. Phase II was the two long corridors that cut the country north-south and east-west. Where those two corridors cross is the single most asked fact in this subject.

- India's road network is the second largest in the world.
- National highways are about 1.46 lakh km (2024-25).
- They are roughly 2 per cent of total road length but carry about 40 per cent of road traffic.
- NHAI was constituted under an Act of 1988 and became operational in 1995.
- Bharatmala Pariyojana (2017) Phase-I covers 34,800 km.

The Golden Quadrilateral circuit, with its highway numbers



ASKED IN RAS

The Golden Quadrilateral has been asked three ways. RAS Pre 2013 - which four cities does it connect? Delhi, Kolkata, Chennai and Mumbai. RAS Pre 2023 - its length? 5,846 km. RAS Pre 2016 - a match-the-following on segment to highway number: Delhi-Kolkata NH-19, Kolkata-Chennai NH-16, Chennai-Mumbai NH-48, Srinagar-Kanniyakumari NH-44.

The two corridors at a glance

Corridor	Route	Mainly which NH
North-South Corridor	Srinagar to Kanniyakumari, with a Salem-Kochi spur	NH-44

Corridor	Route	Mainly which NH
East-West Corridor	Silchar to Porbandar	NH-27
Where they cross	Jhansi, Uttar Pradesh	Both - NHDP Phase II

How to picture the crossing



North-South vs East-West Corridor

North-South Corridor	East-West Corridor
Srinagar (Jammu and Kashmir) to Kanniyakumari (Tamil Nadu)	Silchar (Assam) to Porbandar (Gujarat)
Largely NH-44	Largely NH-27
Has a spur from Salem to Kochi	No spur
NH-44 is India's longest highway, 4,113 km	NH-27 is the second longest, about 3,507 km
Runs top to bottom of the country	Runs from the north-eastern hills to the Arabian Sea

ASKED IN RAS

Asked in RAS Pre 2016 and again in RAS Pre 2021 - where do the North-South and East-West corridors intersect? Answer: Jhansi. In 2021 it came as a three-statement question in which all three statements were correct. If you learn one word from this chapter, learn Jhansi.

ASKED IN RAS

Asked in RAS Pre 2024 - the East-West Corridor connects which two places? Answer: Silchar and Porbandar. Mind the direction of the pair: Silchar is in Assam in the east, Porbandar in Gujarat in the west.

- NH-44 (4,113 km) is the longest national highway in India; NH-27 (about 3,507 km) is second.
- NH-66 runs along the west coast.
- Under the 2010 renumbering, north-south highways get EVEN numbers increasing from east to west.
- East-west highways get ODD numbers increasing from north to south.

YAAD RAKHO

Numbering rule in four words: even goes down, odd goes across. Even numbers are north-south routes, counted westwards from the east coast; odd numbers are east-west routes, counted southwards from the north.

TRAP

Old numbers still float in the options. NH-2 is now NH-19 (Delhi-Kolkata), NH-5 is NH-16 (Kolkata-Chennai), NH-4 and NH-8 are now NH-48, NH-7 is now NH-44. So an option reading 'NH-48 - Delhi to Kolkata' is wrong: NH-48 is the Delhi-Mumbai-Chennai side.

Railways, ports, waterways and airports

The transport tail of the chapter is pure one-liners. Firsts, counts and superlatives - that is all that gets asked. Learn the count of railway zones, the count of major ports and the first three national waterways, and you have covered it.

- India's first passenger train ran from Bori Bunder to Thane, 34 km, on 16 April 1853.
- Indian Railways has 19 zones; Kolkata Metro is counted as one of them.
- The newest zone is South Coast Railway, headquartered at Visakhapatnam.
- India has 12 major ports, administered by the Centre.
- Mundra is a large port but not a 'major port' in the legal sense - it is the standard wrong option.
- Visakhapatnam handles the Bailadila iron ore exports to Japan.

CURRENT LINK

Current link, as of September 2026: the South Coast Railway zone became functional on 1 June 2026 with headquarters at Visakhapatnam and four divisions - Guntakal, Guntur, Vijayawada and Visakhapatnam. It takes Indian Railways to 19 zones. Any 2026 question about 'the newest railway zone' points here.

Port - state - identifier

Port	State	Identifier
Jawaharlal Nehru (Nhava Sheva)	Maharashtra	Largest container-handling major port
Syama Prasad Mookerjee (Kolkata)	West Bengal	Only major riverine port
Kamarajar (Ennore)	Tamil Nadu	First corporatised major port
Deendayal	Gujarat	The renamed Kandla port

Port	State	Identifier
V. O. Chidambaranar	Tamil Nadu	The renamed Tuticorin port
Paradip	Odisha	Bulk cargo, iron ore
Mormugao	Goa	Iron ore exports
Vadhavan (Palghar)	Maharashtra	Approved in 2024 as a new major port

The first three national waterways

Waterway	River / canal	Stretch
NW-1	Ganga	Haldia to Prayagraj, 1,620 km
NW-2	Brahmaputra	Dhubri to Sadiya, 891 km
NW-3	West Coast Canal	Kollam to Kottapuram, 205 km

- The National Waterways Act, 2016 declared III national waterways.
- The Inland Waterways Authority of India was set up in 1986; its headquarters is at Noida, not Delhi.
- NW-1 ends at Prayagraj, not Varanasi - both are single-word substitutions RPSC likes.
- Indira Gandhi International, Delhi, is India's busiest airport.

REVISION RECAP

1. Iron ore - Odisha is the largest producer; belts are Odisha-Jharkhand, Durg-Bastar-Chandrapur (Bailadila), Bellary-Chitradurga (Kudremukh) and Maharashtra-Goa.
2. Manganese - MP leads production (about 31%) but Odisha holds the largest reserves (about 34%); copper works the same way, MP produces and Rajasthan holds the reserves.
3. Bauxite Odisha (Panchpatmali, Koraput); lead-zinc almost entirely Rajasthan (Rampura Agucha, Zawar, Rajpura-Dariba) under Hindustan Zinc; chromite Sukinda; gold Hutti and Kolar.
4. Rock phosphate Jharkhand, gypsum Nagaur-Bikaner, limestone Rajasthan largest producer with Karnataka's largest reserves, diamond Panna, uranium Jaduguda and Tummalapalle.

5. Coal - 96 per cent Gondwana; Jharia largest and coking, Raniganj oldest (1774); Jharkhand largest reserves, Odisha leads production. Rajasthan has lignite, not coal.
6. Lignite - Neyveli largest; Kapurdi-Jalpa and Barsingsar in Rajasthan; Panandhro in Gujarat.
7. Mumbai High (1974, Sagar Samrat) largest oil field; Mangala in the Barmer basin; Digboi (1901) oldest refinery; Jamnagar the world's largest complex; Pachpadra refinery at Balotra.
8. Nuclear - Tarapur oldest (1969), Rawatbhata Canadian-aided, Kudankulam largest and Russian-aided, Kakrapar the first indigenous 700 MWe PHWR.
9. Steel - Bhilai and Bokaro USSR, Rourkela West Germany, Durgapur United Kingdom; Jamshedpur first and private; Visakhapatnam the only shore-based plant.
10. Nicknames - Ahmedabad Manchester of India, Coimbatore Manchester of the South, Chennai Detroit of India, Bengaluru Silicon Valley of India.
11. Eight industrial regions; Hugli is the oldest and jute-based, Chotanagpur is iron and steel, Kollam-Thiruvananthapuram is plantation-based.
12. DFCs - Western Dadri to JNPT (Japanese aid), Eastern Sahnewal to Son Nagar (World Bank); DMIC follows the Western DFC and about 39 per cent of it lies in Rajasthan.
13. Golden Quadrilateral - 5,846 km, NHDP Phase I, completed 2012; Delhi-Kolkata NH-19, Kolkata-Chennai NH-16, Chennai-Mumbai and Mumbai-Delhi NH-48.
14. North-South Corridor Srinagar to Kanniyakumari (NH-44, Salem-Kochi spur); East-West Corridor Silchar to Porbandar (NH-27); the two cross at JHANSI - asked in five papers.
15. NH-44 longest at 4,113 km, NH-27 second at about 3,507 km, NH-66 along the west coast; even numbers run north-south increasing east to west, odd numbers east-west increasing north to south.
16. National highways about 1.46 lakh km, 2 per cent of road length carrying 40 per cent of traffic; NHAI Act 1988, operational 1995; Bharatmala Phase-I 34,800 km.
17. Railways - first train Bori Bunder to Thane, 16 April 1853; 19 zones including Kolkata Metro; South Coast Railway functional 1 June 2026 at Visakhapatnam.
18. 12 major ports - JNPT largest container, Kolkata the only riverine, Kamarajar first corporatised, Deendayal is Kandla, VadHAVAN approved 2024; NW-1 Ganga, NW-2 Brahmaputra, NW-3 West Coast Canal; IWAI at Noida.